

3

Privacy Guard & Redaction

***Owens:** PII/NER detection, deterministic redaction (blur/mask/blackout), redaction validation, canary-test design.*

Role 3 of 6 · PRIVACY

HW Hardware access

Light. Any laptop; occasional GPU time only for tuning a visual face-detector.

You rarely need to book a laptop slot — flag it in advance only if you're doing a visual-detector tuning pass.

P0 Phase 0 — now → 20 Sep 2026 (lock the idea)

- Draft the sensitive-category list (passwords, email/phone/name, government IDs, financial fields, faces, OCR'd text) and the fail-closed decision rule.
- Design the canary-value scheme (e.g. CANARY_EMAIL_12345) that Integration will use for the network-proof demo.

P1 Phase 1 — build the MVP (20 Sep → shortlist)

- Tier 1 — DOM-rule detection (input type / autocomplete / labels / ARIA): fastest, no ML required.
- Tier 2 — lightweight local NER/text-PII model, run only on uncertain regions.
- Tier 3 — visual privacy detection (faces, sensitive imagery), run selectively to control cost.
- Deterministic redaction renderer: blackout / blur / mask / semantic-token — never delegated to an AI model.
- Redaction validator: confirm every flagged region is actually handled before the payload is built.
- Privacy Firewall: the final outbound gate — reject unexpected raw fields, block the request on validation failure.
- Build the privacy inspector UI panel (detected items, redaction status, payload preview) — informational only; the firewall is the real enforcement point.

P2 Phase 2 — rehearse & harden (shortlist → Dec finale)

- Run the canary test end-to-end with Integration: seed values, confirm zero raw canaries reach the server, capture the SHA-256 payload proof for the demo.

DEP Dependencies

YOU DEPEND ON	Perception (DOM/visual capture to run detection on), Extension (the network-intercept hook point).
DEPENDS ON YOU	Server AI (must only ever receive your sanitized output), Integration (dashboard needs your redaction-status feed).

Full team plan, hardware-booking rules, and the master phase timeline: see the companion *PS26171 Team Workflow & Hardware Plan* document.